

# Deep Learning

## Lecture 4: Adversarial models

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## 1 **Generative adversarial networks**

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- properties
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- spectral normalisation
- conditional GANs
- information maximizing GANs
- adversarial autoencoders

## 2 **Popular applications**

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- unpaired translation
- super resolution
- adversarial anomaly detection

## 3 **Adversarial examples**

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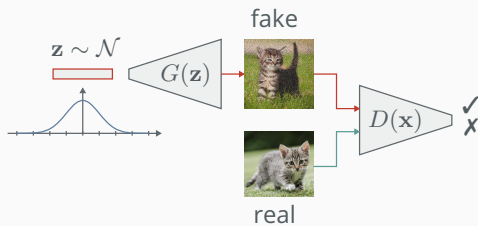
- attacks
- defences

## Definition: generative adversarial networks

A generative adversarial network (GAN) is a non-cooperative zero-sum game where two networks compete against each other [1].

One network  $G(\mathbf{z})$  generates new samples, whereas  $D$  estimates the probability the sample was from the training data rather than  $G$ :

$$\min_G \max_D V(D, G) = \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}(\mathbf{x})} [\log D(\mathbf{x})] \\ + \mathbb{E}_{\mathbf{z} \sim p_{\mathbf{z}}(\mathbf{z})} [\log(1 - D(G(\mathbf{z})))].$$

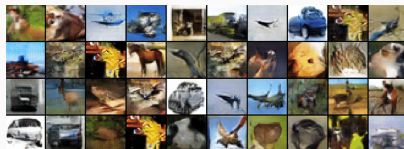
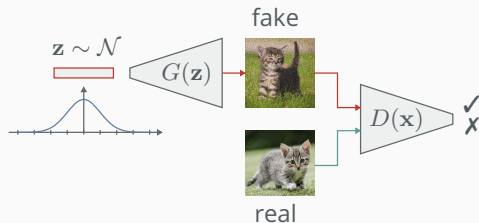


## GAN properties

GANs benefit from differentiable data augmentation [2] for both reals and fakes, but are otherwise notoriously difficult to train:

- Non-convergence
- Diminishing gradient
- Difficult to balance
- Mode collapse (next slide)

[Link to Colab example](#) 



## Definition: mode collapse

This is where the generator rotates through a small subset of outputs, and the discriminator is unable to get out of the trap. Mode collapse is arguably the main limitation of GANs.

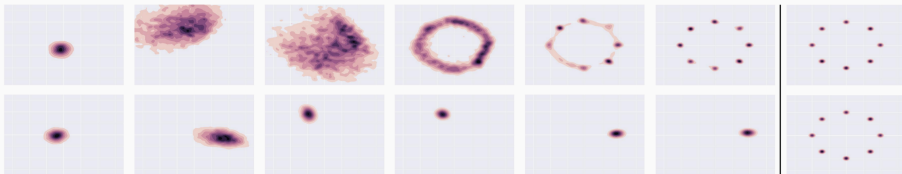


Figure from [3]. The final column shows the target data distribution and the bottom row shows a GAN rotating through the modes.



### Definition: Lipschitz function

A function  $f$  is Lipschitz continuous if it is bounded by how fast it can change. Specifically if there exists a positive real constant  $k$  where:

$$|f(x) - f(y)| \leq k|x - y|,$$

for all  $y$  sufficiently near  $x$ . For example, any function with a bounded first derivative is a Lipschitz function.

Wasserstein GANs [4, 5] were the first to reduce mode collapse in GANs by lowering the Lipschitz constant for the discriminator function.

### Distance functions are 1-Lipschitz





## Definition: spectral normalisation

The matrix (spectral) norm defines how much a matrix can stretch a vector  $\mathbf{x}$ :

$$\|\mathbf{A}\| = \max_{\mathbf{x} \neq 0} \frac{\|\mathbf{Ax}\|}{\|\mathbf{x}\|}$$

Spectral norm [6] normalises the weights for each layer using the spectral norm  $\sigma(\mathbf{W})$  such that the Lipschitz constant for every layer and the whole network is 1:

$$\hat{\mathbf{W}}_{\text{SN}} = \mathbf{W} / \sigma(\mathbf{W})$$

$$\sigma(\hat{\mathbf{W}}_{\text{SN}}(\mathbf{W})) = 1$$

$$\|f\|_{\text{Lip}} = 1$$

## Pseudocode: 1-Lipschitz discriminator

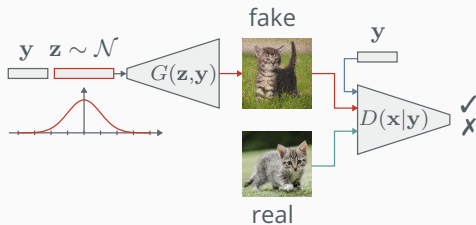
```
class Discriminator(nn.Module):
    def __init__(self, f=64):
        super().__init__()
        self.discriminate = nn.Sequential(
            spectral_norm(Conv2d(1, f, 3, 1, 1)),
            nn.LeakyReLU(0.1, inplace=True),
            nn.MaxPool2d(kernel_size=(2,2)),
            spectral_norm(Conv2d(f, f*2, 3, 1, 1)),
            nn.LeakyReLU(0.1, inplace=True),
            nn.MaxPool2d(kernel_size=(2,2)),
            spectral_norm(Conv2d(f*2, f*4, 3, 1, 1)),
            nn.LeakyReLU(0.1, inplace=True),
            nn.MaxPool2d(kernel_size=(2,2)),
            spectral_norm(Conv2d(f*4, f*8, 3, 1, 1)),
            nn.LeakyReLU(0.1, inplace=True),
            nn.MaxPool2d(kernel_size=(2,2)),
            spectral_norm(Conv2d(f*8, 1, 3, 1, 1)),
            nn.Sigmoid()
        )
```

## Definition: conditional GAN

GANs can be conditioned with labels  $\mathbf{y}$  if available [7] by feeding the label information into both the generator and the discriminator:

$$\min_G \max_D V(D, G) = \mathbb{E}_{\mathbf{x}, \mathbf{y} \sim p_{\text{data}}(\mathbf{x})} [\log D(\mathbf{x}|\mathbf{y})] \\ + \mathbb{E}_{\mathbf{z} \sim p_{\mathbf{z}}(\mathbf{z})} [\log(1 - D(G(\mathbf{z}, \mathbf{y})|\mathbf{y}))].$$

[Link to Colab example](#) 



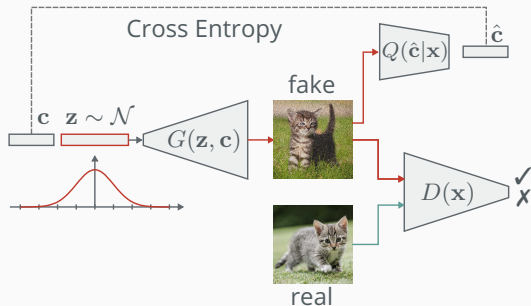
### Definition: information maximizing GANs

GANs can be trained to learn disentangled latent representations in a completely unsupervised manner. InfoGAN [8] popularised this by maximizing mutual information between the observation and a subset of the latents:

$$\min_{G,Q} \max_D V_{\text{InfoGAN}}(D, G, Q) = V(D, G) - \lambda L_I(G, Q)$$

where  $L_I(G, Q)$  is a variational lower bound of the mutual information.

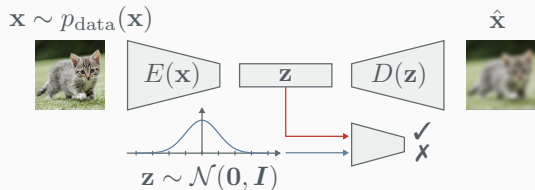
[Link to Colab example](#) 



## Definition: adversarial autoencoders

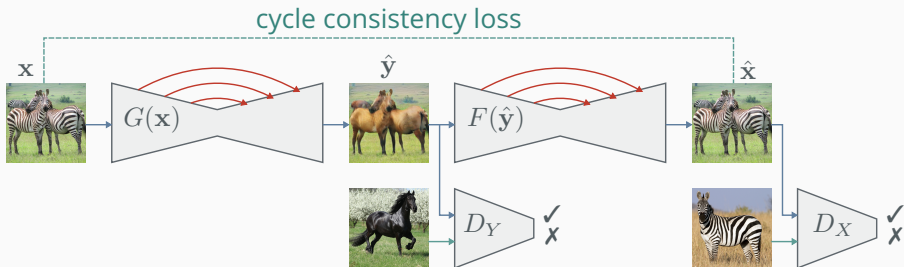
Adversarial autoencoders [9] are generative models that permit sampling.

In addition to the reconstruction loss, such  $\|\mathbf{x} - \hat{\mathbf{x}}\|^2$ , they use adversarial training to match the aggregated posterior of the hidden code vector  $\mathbf{z}$  of the autoencoder with an arbitrary prior distribution, such as  $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ .



## Definition: unpaired translation (CycleGAN)

CycleGAN [10] propose an adversarial architecture that enables unpaired image translation. It has twin residual generators and two discriminators, which translate between the domains, alongside a cycle consistency loss (an L1 norm) which ensures the mapping can recover the original image.

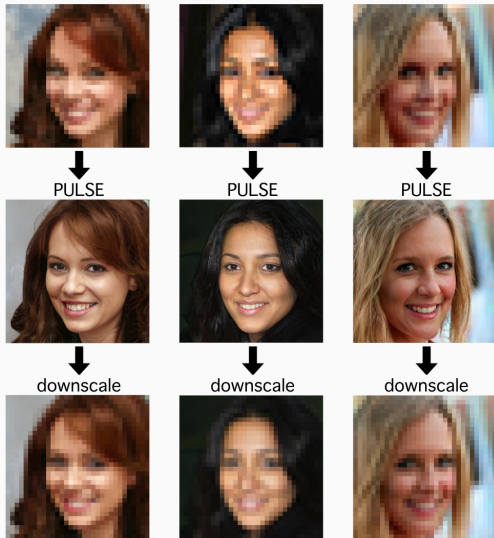


## Definition: super-resolution

Adversarial models are popular in super-resolution approaches. The challenge is that a single low-resolution (LR) input can map to a distribution of high-resolution (HR) outputs.

PULSE [11] investigates this by projecting points in the search of the latent space of StyleGAN (a large conditional GAN) onto a hypersphere, which ensures probable outputs in the high-dimensional latent space.

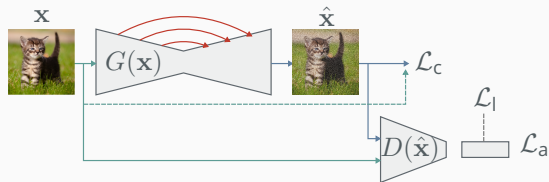
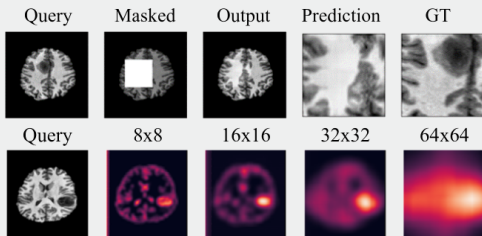
**Online example** [↗](#)



# Popular applications adversarial anomaly detection

## Definition: anomaly detection

Unsupervised anomaly detectors [12] learn a normal distribution over (healthy) observations. Then, when they observe something not observed in training (unhealthy/dangerous), they fail to reconstruct - detecting it as an anomaly. Region-based anomaly detectors [13] learn a distribution over inpainted (erased) regions.



## Definition: adversarial examples

These are small but intentionally worst-case perturbations that fool the model to output incorrect answers with high confidence [14]. It is possible to generate examples that also fool the human visual system [15]. Cat or dog?

original



adv



## Example: adversarial examples

Example adding an imperceptibly small vector by the sign of the elements of the gradient of the cost function with respect to the input [14]:

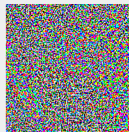


$x$

“panda”

57.7% confidence

+ .007 ×

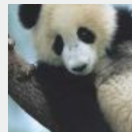


$\text{sign}(\nabla_x J(\theta, x, y))$

“nematode”

8.2% confidence

=



$x +$

$\epsilon \text{sign}(\nabla_x J(\theta, x, y))$

“gibbon”

99.3 % confidence

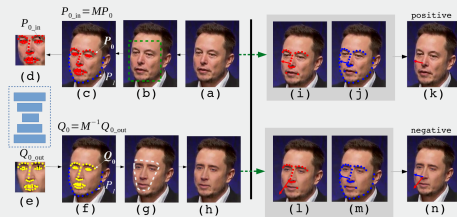
## Definition: adversarial defence

There are several defence strategies that introduce the adversarial examples into training [14]. A popular approach uses U-Net to denoise and reduce the amplification of the adversarial perturbations [16].

Black-box adversarial defence is where an adversary can only monitor the outputs of the model. White-box methods are more difficult, as an adversary has access to the model allowing for specific attacks. White-box defence generally overfits to the attack used during training.

## Example: adversarial defences

Question: What is the behaviour at the limit of the adversarial generative model arms-race? Who wins at convergence?





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